

Python Practice:

1) # This program is written to add two numbers

Start the program

num1 = 2

num2 = 6

print("\t\tWelcome to the Python Program!!\n")

Display the above text

Add the two numbers & display the sum

sum = num1 + num2

print(f"The Calculated sum is = {sum}")

End of the program

2) # This program is written to calculate the area & circumference of a circle

Start the program

pi = 3.1416

print("\t\tWelcome to the Python Program!!\n")

Display the above text

Get the radius of the circle from the user

radius = float(input("Enter the radius of a circle: "))

Calculate the area & display the result

area = pi * radius * radius

print(f"The Area of a circle is = {area:.3f}")

Calculate the circumference & display the result

circumference = 2 * pi * radius

print(f"The Circumference of a circle is = {circumference:.3f}")

End of the program

3) # This Python program is made to calculate simple and compound interest

import math

Start of the program

print("\t\tWelcome to the Simple & Compound Calculator Program!!\n")

Display the welcome message

Take 3 user inputs

P = float(input("Enter your principal amount: "))

r = float(input("Enter your interest rate: "))

n = float(input("Enter the time (in years): "))

Assigning formulas to variables

$I = P * r * n / 100$

$p = 1 + (r / 100)$

$C = P * \text{math.pow}(p, n)$

Calculate & display the results

print(f"The result of Simple Interest is = {I:.2f} Taka")

print(f"The result of Compound Interest is = {C:.2f} Taka")

End of the program

4) # This Python program is made to swap values of two variables without using a third variable

Start of the program

x = 10

y = 20

Declare & assign variables

print("\t\t\tWelcome to the Python Program!!\n")

Display the welcome message

Swap the values without using a third variable

$x = x + y$

$y = x - y$

$x = x - y$

Display the result

print(f"The swapped value of x is = {x}")

print(f"The swapped value of y is = {y}")

End of the program

5) # This Python program is made to swap values of two variables using a third variable

Start of the program

x = 10

y = 20

Declare & assign variables

print("\t\t\tWelcome to the Python Program!!\n")

Display the welcome message

Swap the values using a third variable

z = x

x = y

y = z

Display the swapped values of the 2 variables

```
print(f"The swapped value of x is = {x}")
```

```
print(f"The swapped value of y is = {y}")
```

End of the program

6) # This Python program is made to display the size of every data type using the "sys.getsizeof" function.

```
import sys
```

Start of the program

```
x = 0
```

```
y = 0.0
```

```
z = 0.0
```

```
a = 'a'
```

Declare & assign variables

```
p = sys.getsizeof(x)
```

```
q = sys.getsizeof(y)
```

```
r = sys.getsizeof(z)
```

```
s = sys.getsizeof(a)
```

```
print("\t\t\tWelcome to the Python Program!!\n")
```

Display the welcome message

Display the size of the data types

```
print(f"The size of int data type is = {p} bytes")
```

```
print(f"The size of float data type is = {q} bytes")
```

```
print(f"The size of double data type is = {r} bytes")
```

```
print(f"The size of char data type is = {s} bytes")
```

End of the program

7) # This Python program is made to find the largest of three numbers using conditional expressions

Start of the program

```
print("\t\t\tWelcome to the Python Program!!\n")
```

Display the welcome message

Take 3 user inputs

```
x = float(input("Enter the first number: "))
```

```
y = float(input("Enter the second number: "))
```

```
z = float(input("Enter the third number: "))
```

Use of conditional expressions (ternary operators)

```
largest = x if x >= y and x >= z else (y if y >= x and y >= z else z)
```

Display the largest number

```
print(f"The largest number is = {largest:.2f}")
```

End of the program

8) # This Python program is made to illustrate the use of unary prefix and postfix increment and decrement operators.

Start of the program

```
x = 100
```

Declare & assign variables

```
print("\t\t\tWelcome to the Python Program!!\n")
```

Display the welcome message

```
print(f"The value of x is = {x}")
```

Python does not have built-in prefix or postfix operators, so we'll simulate the behavior

Prefix increment

```
x += 1
```

```
print(f"The prefix increment value of x is = {x}")
```

Reset x to 100

```
x = 100
```

Postfix increment (simulate by printing before incrementing)

```
print(f"The postfix increment value of x is = {x}")
```

```
x += 1
```

Reset x to 100

```
x = 100
```

Prefix decrement

```
x -= 1
```

```
print(f"The prefix decrement value of x is = {x}")
```

Reset x to 100

```
x = 100
```

Postfix decrement (simulate by printing before decrementing)

```
print(f"The postfix decrement value of x is = {x}")
```

```
x -= 1
```

End of the program

9) # This Python program calculates the area of a rectangle

Start of the program

```
print("Enter a numeric value of length: ")
```

```
l = int(input())
```

```
print("Enter a numeric value of width: ")
```

```
w = int(input())
```

```
# Calculate the area
```

```
v = l * w
```

```
# Display the result
```

```
print(f"The Calculated Area is = {v} m^2")
```

```
# End of the program
```

10) # This Python program calculates a problem involving exponentiation and square root.

```
import math
```

```
# Start of the program
```

```
print("Enter 2 numbers: ")
```

```
a = float(input())
```

```
b = float(input())
```

```
# Calculate powers and square root
```

```
x = math.pow(a, 3)
```

```
y = math.pow(b, 5)
```

```
z = x + y
```

```
w = math.sqrt(z)
```

```
# Display the result
```

```
print(f"The Calculated Result is = {w:.2f}")
```

```
# End of the program
```

11) # This Python program displays the size of variables 'a' and 'b'

```
import sys
```

```
# Start of the program
```

```
a = 0
```

```
b = 0
```

```
x = sys.getsizeof(a)
```

```
print(f"The size of int a is = {x}")
```

```
y = sys.getsizeof(b)
```

```
print(f"The size of int b is = {y}")
```

```
# End of the program
```

12) # This Python program displays the increment value of a and the value of b

```
# Start of the program
```

```
print("Enter a number: ")
```

```

a = int(input())

b = a

a += 1

# Display the increment value of a and the value of b

print(f"The increment value of a is = {a}")

print(f"The value of b is = {b}")

# End of the program

```

13) # This Python program determines if a number is even or odd

Start of the program

```

print("Enter a number: ")

x = int(input())

y = x % 2

if y == 0: print("The number is even.")
else: print("The number is odd.")

# End of the program

```

14) # This Python program determines if a number is positive, negative, or zero

Start of the program

```

print("Enter a number: ")

x = int(input())

if x > 0: print("The number is positive.")
elif x < 0: print("The number is negative.")
else: print("The number is zero.")

# End of the program

```

15) # This Python program checks whether the entered year is a leap year or not

Start of the program

```

print("\t\t\tWelcome to the Python Program!!\n")

# Display the welcome message

# Take a year from user as input

print("Enter a year: ")

year = int(input())

# Check if the year is a leap year

if (year % 400 == 0): print("This year is a Leap Year.")

elif (year % 100 != 0 and year % 4 == 0):

    print("This year is a Leap Year.")

```

```
else: print("This year is not a leap year.")
```

```
# End of the program
```

16) # This Python program determines whether the input character is capital, small letter, digit, or special symbol

```
# Start of the program
```

```
print("\t\t\tWelcome to the Python Program!!\n")
```

```
# Display the welcome message
```

```
# Take 1 letter from user as input
```

```
print("Enter a letter: ")
```

```
x = input()
```

```
# Check the type of the input character
```

```
if 'A' <= x <= 'Z': print("This letter is Capital.")
```

```
elif 'a' <= x <= 'z': print("This letter is Small.")
```

```
elif '0' <= x <= '9': print("This letter is a Digit.")
```

```
else: print("This letter is a Special Symbol.")
```

```
# End of the program
```

17) # This Python program compares two numbers and prints '>', '<', or '='

```
# Start of the program
```

```
print("Enter 2 numbers: ")
```

```
x = int(input())
```

```
y = int(input())
```

```
# Compare the numbers and print the result
```

```
if x > y: print(">")
```

```
elif x < y: print("<")
```

```
else: print("=")
```

```
# End of the program
```

18) # This Python program computes the grade of students using if-elif-else ladder

```
# Start of the program
```

```
print("\t\t\tWelcome to the Python Program!!\n")
```

```
# Display the welcome message
```

```
# Take marks from user as input
```

```
print("Enter the marks: ")
```

```
Marks = int(input())
```

```
# Determine the grade based on marks
```

```
if 90 <= Marks <= 100: print("The Grade is: A+\n")
```

```
elif 80 <= Marks < 90: print("The Grade is: A\n")
elif 70 <= Marks < 80: print("The Grade is: B+\n")
elif 60 <= Marks < 70: print("The Grade is: B\n")
elif 50 <= Marks < 60: print("The Grade is: C\n")
else: print("The Grade is: F\n")

# End of the program
```

19) # Simple Calculator Program in Python

```
# Start of the program

print("\t\t\tWelcome to the Simple Calculator Program!!\n")

# Display the welcome message

# Take input from the user for two numbers

print("Enter first number: ")

num1 = float(input()) # Convert input to float for decimal operations

print("Enter second number: ")

num2 = float(input()) # Convert input to float for decimal operations

# Menu for operations

print("\nSelect operation:")

print("1. Addition")

print("2. Subtraction")

print("3. Multiplication")

print("4. Division")

# Take input from the user for operation choice

choice = int(input("Enter choice (1/2/3/4): "))

# Perform operation based on user's choice

if choice == 1:

    result = num1 + num2

    print(f"{num1} + {num2} = {result}")

elif choice == 2:

    result = num1 - num2

    print(f"{num1} - {num2} = {result}")

elif choice == 3:

    result = num1 * num2

    print(f"{num1} * {num2} = {result}")

elif choice == 4:
```



```

if num2 == 0:

    print("Error! Division by zero is not allowed.")

else:

    result = num1 / num2

    print(f"{num1} / {num2} = {result}")

else:

    print("Invalid input! Please choose a valid operation (1/2/3/4).")

# End of the program

```

20) # Python program to check divisibility of a number and print messages

```

# Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

```

```

# Display the welcome message

```

```

# Take input from user for a number

```

```

print("Enter a number: ")

```

```

x = int(input())

```

```

# Check divisibility and print messages

```

```

if x % 2 == 0: print("The number is even.")

```

```

if x % 3 == 0: print("prop")

```

```

if x % 4 == 0: print("Tess")

```

```

if x % 5 == 0: print("penta")

```

```

# End of the program

```

21) # Python program to check number properties

```

# Start of the program

```

```

print("\t\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

```

```

# Display the welcome message

```

```

# Take input from user for a number

```

```

print("Enter a number: ")

```

```

x = int(input())

```

```

# Check if the number is even or odd

```

```

if x % 2 == 0: print("The number is even.")

```

```

else: print("The number is odd.")

```

```

# Check divisibility and print messages

```

```

if x % 3 == 0: print("prop")

```

```

else: print("Not prop")

```

```
if x % 4 == 0: print("Tess")
else: print("Not Tess")
if x % 5 == 0: print("penta")
else: print("Not penta")
# End of the program
```

22) # Python program to check character type and provide additional information

```
# Start of the program
print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")
# Display the welcome message
# Take input from user for a character
print("Enter a character: ")
x = input()[0] # Take the first character of the input
# Check if the character is an alphabet (uppercase or lowercase)
if (x >= 'A' and x <= 'Z') or (x >= 'a' and x <= 'z'):
    print("The character is an alphabet.")
    if x >= 'A' and x <= 'Z':
        print("The character is an uppercase letter.")
    elif x >= 'a' and x <= 'z':
        print("The character is a lowercase letter.")
else:
    # Check if the character is a digit
    if x >= '0' and x <= '9':
        print("The character is a digit.")
    if x < '5':
        print("X is less than 5.")
    else:
        print("X is greater than or equal to 5.")
else:
    print("The character is a special symbol.")
# End of the program
```

23) # Python program to calculate square root until a negative number is entered

```
import math
# Start of the program
print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")
```

```

while True:

    # Take input from user for a number

    print("Enter a number: ")

    y = float(input())

    # Check if the number is negative to break the loop

    if y < 0:

        break

    # Calculate square root and display the result

    z = math.sqrt(y)

    print(f"The square root of {y:.0f} is: {z:.3f}")

# End of the program

```

24) # Python program to count the number of digits in a number

```

# Start of the program

print("\t\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

# Take input from user for a number

print("Enter a number: ")

x = int(input())

# Initialize variables

count = 0

# Loop to count digits

while True:

    z = x % 10 # Get the last digit

    x = x // 10 # Remove the last digit

    count += 1 # Increment digit count

    if x == 0:

        break # Exit loop when no digits are left

# Print the number of digits

print(f"The number of digits is = {count}")

# End of the program

```

25) # Python program to print a triangular pattern of asterisks

```

# Start of the program

print("\t\t\tWELCOME TO THE DRAGON WARRIOR'S PROGRAM!!\n")

# Loop to print the triangular pattern

for x in range(1, 11): # Outer loop for rows from 1 to 10

```

```

    for y in range(1, x + 1): # Inner loop for columns, prints * x times
        print(" *", end="") # Print asterisk with space
    print() # Move to the next line after each row
# End of the program

26) # Python program to find the greatest common divisor (GCD) of two numbers

# Start of the program
print("\t\tWELCOME TO THE DRAGON WARRIOR'S PROGRAM!!\n")

# Initialize variables
a = 10
b = 18

# Determine the smaller number between a and b for efficient loop
if a > b: smaller = b
else: smaller = a

gcd = 1 # Initialize gcd to 1

# Loop to find the greatest common divisor (GCD)
for x in range(1, smaller + 1):
    if a % x == 0 and b % x == 0:
        gcd = x # Update gcd when x divides both a and b

# Print the greatest common divisor (GCD)
print(f"The greatest common divisor (GCD) of {a} and {b} is: {gcd}")

# End of the program

27) # Python program to print a triangular pattern with "Arafat"

# Start of the program
print("\t\tWELCOME TO THE DRAGON WARRIOR'S PROGRAM!!\n")

# Initialize variables
z = 9 # Number of rows

# Loop to print the triangular pattern
for x in range(1, z + 1): # Outer loop for rows from 1 to z
    for y in range(0, 2 * (z - x) - 1): # Loop for spaces
        print(" ", end="") # Print two spaces for each iteration
    for y in range(1, x + 1): # Loop to print "Arafat"
        print(" Arafat", end="") # Print " Arafat"
    print() # Move to the next line after each row

# End of the program

```

28) # Python program to calculate the sum of 10 numbers entered by the user

```
print("\t\t\tWELCOME TO THE PYTHON PROGRAM!!\n") # Display the welcome message
```

```
sum = 0 # Initialize sum variable to store the cumulative sum
```

```
print("Enter 10 numbers:")
```

```
for i in range(1, 11):
```

```
    x = int(input(f"Enter number {i}: ")) # Take input from the user
```

```
    sum += x # Add the current input to the sum
```

```
print(f"The sum of the 10 numbers is: {sum}") # Display the sum
```

```
# End of the program
```

29) # Python program to print the multiplication table of a number entered by the user

```
print("\t\t\tWELCOME TO THE PYTHON PROGRAM!!\n") # Display the welcome message
```

```
Num = int(input("Enter a number: ")) # Take input from user
```

```
print(f"The multiplication table of {Num} is = ") # Display the message
```

```
for x in range(1, 11): print(f" {Num} X {x} = {Num * x}") # Print each multiplication line
```

```
# End of the program
```

30) # Python program to print a pattern of asterisks

```
print("\t\t\tWELCOME TO THE PYTHON PROGRAM!!\n") # Display the welcome message
```

```
for x in range(1, 5): # Outer loop to iterate 4 times
```

```
    for y in range(1, 7): # Inner loop to print 6 asterisks in each row
```

```
        print(" *", end="") # Print an asterisk with a space
```

```
    print() # Move to the next line after printing 6 asterisks
```

```
# End of the program
```

31) # Start of the program

```
# Display the welcome message
```

```
print("\t\t\tWELCOME TO THE PYTHON PROGRAM!!\n")
```

```
# Declare variables
```

```
x = int(input("Enter a 3 digit number: ")) # Take an input from the user
```

```
RealNum = x
```

```
ReverseNum = 0
```

```
# Calculate the reverse number
```

```
while x > 0:
```

```
    Remainder = x % 10
```

```
    ReverseNum = ReverseNum * 10 + Remainder
```

```
    x = x // 10
```

```
# Check if the given number is a palindrome or not

if RealNum == ReverseNum: print(f"{RealNum} is a Palindrome.")
else: print(f"{RealNum} is not a Palindrome.")

# End of the program
```

32) # Start of the program

```
# Display the welcome message

print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

# Take input from the user

a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))

# Calculate the GCD of the numbers

for i in range(1, min(a, b) + 1):
    if a % i == 0 and b % i == 0:
        gcd = i
```

```
# Calculate the LCM of the numbers
```

```
lcm = (a * b) // gcd
```

```
# Display the results
```

```
print(f"The GCD of {a} and {b} is = {gcd}")
```

```
print(f"The LCM of {a} and {b} is = {lcm}")
```

```
# End of the program
```

33) print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

```
# Declare a list to store the numbers
```

```
numbers = [0] * 10
```

```
# Take input from the user
```

```
print("Enter 10 numbers: ")
```

```
for i in range(10):
```

```
    numbers[i] = int(input())
```

```
# Print the numbers
```

```
for num in numbers:
```

```
    print(num, end=" ")
```

```
# End of the program
```

34) # Start of the program

```
print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")
```

```
# Declare a list to store the numbers with fixed size
```

```

numbers = [0] * 5

sum = 0

# Take input from the user
print("Enter 5 positive numbers: ")

for i in range(5):

    numbers[i] = int(input())

# Calculate the sum
for num in numbers:

    sum += num

# Print the result
print(f"The calculated result is = {sum}")

35) # Start of the program
print("\t\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

# Declare variables

flag = 0

key = 3

x = [2, 3, 4, 6, 9]

# Search for the key in the list

for i in range(5):

    if x[i] == key:

        flag = 1

        break # Break the loop if the key is found

# Check if the key was found and print the result
if flag == 0: print("Not Found")

else: print("Found")

36) mat1 = [[0 for _ in range(3)] for _ in range(2)]

mat2 = [[0 for _ in range(3)] for _ in range(2)]

sum_matrix = [[0 for _ in range(3)] for _ in range(2)]

print("Enter Matrix 1 Data:")

for i in range(2):

    for j in range(3):

        mat1[i][j] = int(input(f"Element [{i}][{j}]: "))

print("Enter Matrix 2 Data:")

for i in range(2):

```

```

for j in range(3):
    mat2[i][j] = int(input(f"Element [{i}][{j}]: "))

for i in range(2):
    for j in range(3):
        sum_matrix[i][j] = mat1[i][j] + mat2[i][j]
print("Sum of Matrix 1 and Matrix 2 is:")

for i in range(2):
    for j in range(3):
        print(sum_matrix[i][j], end=" ")
    print()

```

```

37) name = input("Enter your name: ")
print(name)
count = 0
for char in name:
    count += 1
print("\nLength of the string:", count)

```

```

38) name = input("Enter your name: ")
print(name)
count = len(name)
print(count)

```

```

39) name = input("Enter a string: ")
a = 0 # count of alphabets
d = 0 # count of digits
s = 0 # count of special characters

for char in name:
    if char.isalpha():
        a += 1
    elif char.isdigit():
        d += 1
    else:
        s += 1

print(f"No of alphabets= {a}")
print(f"No of digits= {d}")
print(f"No of special characters= {s}")

```


40) # Start of the program

```
KING = input("Enter a string: ")
```

```
# Declare variables
```

```
count = 0
```

```
# Loop through each character in the string to count its length
```

```
for char in KING:
```

```
    count += 1
```

```
# Display the length of the string
```

```
print("\nThe Length of the string:", count)
```

```
# End of the program
```

41) # Start of the program

```
print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")
```

```
# Take an input from the user
```

```
KING = input("Enter a string: ")
```

```
# Declare variables
```

```
alpha = 0
```

```
digit = 0
```

```
special = 0
```

```
# Loop through each character in the string
```

```
for char in KING:
```

```
    if char.isalpha():
```

```
        alpha += 1
```

```
    elif char.isdigit():
```

```
        digit += 1
```

```
    else:
```

```
        special += 1
```

```
# Display the count of alphabets, digits, and special characters
```

```
print("\nThe Number of alphabets =", alpha)
```

```
print("The Number of digits =", digit)
```

```
print("The Number of special characters =", special)
```

```
# End of the program
```

42) # Start of the program

```
import sys
```

```
x = "Arafat"
```

```

y = " Sakib"

# Display the welcome message

print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

# Compare the two strings

if x == y:

    result = 0

elif x < y:

    result = -1

else:

    result = 1

# Display the string compare result

print(f"The 2 strings compared result is: {result}")

# End of the program

```

43) # Start of the program

```

x = "Arafat "
y = "Sakib."

# Display the welcome message

print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

# Concatenate the two strings

concatenated_string = x + y

# Display the concatenated string

print(f"The concatenated string is: {concatenated_string}")

# End of the program

```

44) # Start of the program

```

# Display the welcome message

print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

# Take an input from the user

Num = int(input("Enter a number: "))

# Check divisibility

if Num % 5 == 0 and Num % 11 == 0:

    print("The number is divisible by both 5 & 11")

elif Num % 5 == 0 and Num % 11 != 0:

    print("The number is divisible by 5 but not 11")

elif Num % 11 == 0 and Num % 5 != 0:

```

```

    print("The number is divisible by 11 but not 5")
else:
    print("The number is not divisible by both 5 & 11")
# End of the program

45) # Start of the program
# Display the welcome message
print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")
# Take an input from the user
StringA = input("Enter a string: ")
# Copy the first 5 characters from StringA to StringB
StringB = StringA[:5]
# Display the copied string
print(f"The Copied String B Is: {StringB}")
# End of the program

46) # Start of the program
# Display the welcome message
print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")
# Take an input from the user
KING = input("Enter a string: ")
# Calculate the length of the string
y = len(KING)
# Display the characters in reverse form
print("The Characters Of Reverse Form Is:")
for x in range(y - 1, -1, -1):
    print(KING[x])

47) # Display the welcome message
print("\t\tWelcome to the Python Program!!\n")
# Take two numbers as input from the user
print("Enter 2 numbers:")
x = int(input("First number: "))
y = int(input("Second number: "))
# Define the functions for arithmetic operations
def add(a, b): return a + b
def sub(a, b): return a - b

```

```

def mul(a, b): return a * b

def div(a, b): return a // b # Use integer division to match C behavior

# Perform arithmetic operations and print results

result = add(x, y)

print(f"The addition result: {result}")

res = sub(x, y)

print(f"The subtraction result: {res}")

re = mul(x, y)

print(f"The multiplication result: {re}")

# Check for division by zero

if y != 0:

    r = div(x, y)

    print(f"The division result: {r}")

else: print("Division by zero is not allowed.")

48) # Define the function to check if a number is even

def is_even(n):

    return n % 2 == 0

# Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

# Take an input from the user

num = int(input("Enter a number: "))

# Check if the number is even or odd and print the result

if is_even(num):

    print(f"The given number {num} is even.")

else:

    print(f"The given number {num} is odd.")

49) # Define the function to calculate the sum of the first num natural numbers

def sum_natural_numbers(num):

    total = 0

    for i in range(1, num + 1):

        total += i

    return total

# Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

```

```
# Initialize the number
```

```
num = 20
```

```
# Calculate the sum using the function
```

```
Add = sum_natural_numbers(num)
```

```
# Print the result
```

```
print(f"The sum of the first {num} natural numbers is: {Add}")
```

```
50) # Define the function to increment the values of the variables
```

```
def incre(x, y, z):
```

```
    return x + 1, y + 1, z + 1
```

```
# Start of the program
```

```
print("\t\t\tWelcome to the Python Program!!\n")
```

```
# Take input from the user
```

```
a = int(input("Enter a number for a: "))
```

```
b = int(input("Enter a number for b: "))
```

```
c = int(input("Enter a number for c: "))
```

```
# Display original numbers
```

```
print("\nOriginal numbers:")
```

```
print(f"a = {a}")
```

```
print(f"b = {b}")
```

```
print(f"c = {c}")
```

```
# Modify the numbers using the function
```

```
a, b, c = incre(a, b, c)
```

```
# Display modified numbers
```

```
print("\nModified numbers:")
```

```
print(f"a = {a}")
```

```
print(f"b = {b}")
```

```
print(f"c = {c}")
```

```
51) # Define the function to swap values by value (doesn't actually change the original values)
```

```
def swap1(a, b):
```

```
    a, b = b, a
```

```
    print(f"Inside swap by value: a = {a}, b = {b}")
```

```
    return a, b
```

```
# Define the function to swap values by reference
```

```
def swap2(lst):
```

```

lst[0], lst[1] = lst[1], lst[0]

print(f"Inside swap by reference: a = {lst[0]}, b = {lst[1]}")

# Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

# Initialize variables

x = 10

y = 20

# Display before swap by value

print(f"Before swap by value: x = {x}, y = {y}")

# Swap by value

x, y = swap1(x, y)

# Display after swap by value

print(f"After swap by value: x = {x}, y = {y}")

# Display before swap by reference

print(f"Before swap by reference: x = {x}, y = {y}")

# Swap by reference

swap2([x, y])

# Display after swap by reference

print(f"After swap by reference: x = {x}, y = {y}")

52) # Define a class to simulate the structure

class Company:

    def __init__(self, name="", address="", phone="", no_of_employees=0):

        self.name = name

        self.address = address

        self.phone = phone

        self.no_of_employees = no_of_employees

# Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

# Create an instance of the Company class

my_company = Company()

# Take inputs from the user

my_company.name = input("Enter the name of the company: ").strip()

my_company.address = input("Enter the address of the company: ").strip()

my_company.phone = input("Enter the phone number of the company: ").strip()

```

```

my_company.no_of_employees = int(input("Enter the number of employees in the company: "))

# Display the company information

print("\nCompany Information:")

print(f"Name: {my_company.name}")

print(f"Address: {my_company.address}")

print(f"Phone: {my_company.phone}")

print(f"Number of Employees: {my_company.no_of_employees}")

```

53) # Define a class to represent a complex number

```

class Complex:

```

```

    def __init__(self, real=0.0, imag=0.0):

```

```

        self.real = real

```

```

        self.imag = imag

```

```

# Function to add two complex numbers

```

```

def add_complex(a, b):

```

```

    return Complex(a.real + b.real, a.imag + b.imag)

```

```

# Function to subtract two complex numbers

```

```

def subtract_complex(a, b):

```

```

    return Complex(a.real - b.real, a.imag - b.imag)

```

```

# Function to display a complex number

```

```

def display_complex(c):

```

```

    if c.imag >= 0:

```

```

        print(f"{c.real:.2f} + {c.imag:.2f}i")

```

```

    else:

```

```

        print(f"{c.real:.2f} - {abs(c.imag):.2f}i")

```

```

# Start of the program

```

```

print("\t\t\tWelcome to the Python Program!!\n")

```

```

# Take input for the first complex number

```

```

real1 = float(input("Enter the real part of the first complex number: "))

```

```

imag1 = float(input("Enter the imaginary part of the first complex number: "))

```

```

num1 = Complex(real1, imag1)

```

```

# Take input for the second complex number

```

```

real2 = float(input("Enter the real part of the second complex number: "))

```

```

imag2 = float(input("Enter the imaginary part of the second complex number: "))

```

```

num2 = Complex(real2, imag2)

```

```

# Perform addition and subtraction

sum_result = add_complex(num1, num2)

difference_result = subtract_complex(num1, num2)

# Display the results

print("\nThe sum of the complex numbers is: ", end="")

display_complex(sum_result)

print("The difference of the complex numbers is: ", end="")

display_complex(difference_result)

```

54) class Student:

```

def __init__(self, roll_no=0, name="", address="", age=0, average_marks=0.0):

    self.roll_no = roll_no

    self.name = name

    self.address = address

    self.age = age

    self.average_marks = average_marks

def read_stu_det(students):

    for i in range(len(students)):

        print(f"Enter details for student {i + 1}:")

        students[i].roll_no = int(input("Roll No: "))

        students[i].name = input("Name: ")

        students[i].address = input("Address: ")

        students[i].age = int(input("Age: "))

        students[i].average_marks = float(input("Average Marks: "))

        print()

def dis_stu_det(students):

    print("Displaying student details:\n")

    for i, student in enumerate(students):

        print(f"Student {i + 1}:")

        print(f"Roll No: {student.roll_no}")

        print(f"Name: {student.name}")

        print(f"Address: {student.address}")

        print(f"Age: {student.age}")

        print(f"Average Marks: {student.average_marks:.2f}")

        print()

```



```

# Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

# Create a list of Student objects

students = [Student() for _ in range(12)]

# Read student details

read_stu_det(students)

# Display student details

dis_stu_det(students)

55) class Student:

    def __init__(self, roll_no=0, name="", address="", age=0, average_marks=0.0):

        self.roll_no = roll_no

        self.name = name

        self.address = address

        self.age = age

        self.average_marks = average_marks

def read_stu_det(student):

    print("Enter details for the student:")

    student.roll_no = int(input("Roll No: "))

    student.name = input("Name: ")

    student.address = input("Address: ")

    student.age = int(input("Age: "))

    student.average_marks = float(input("Average Marks: "))

    print()

def dis_stu_det(student):

    print("Displaying student details:\n")

    print(f"Student:")

    print(f"Roll No: {student.roll_no}")

    print(f"Name: {student.name}")

    print(f"Address: {student.address}")

    print(f"Age: {student.age}")

    print(f"Average Marks: {student.average_marks:.2f}")

    print()

# Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

```

```
# Create a Student object
```

```
student = Student()
```

```
# Read student details
```

```
read_stu_det(student)
```

```
# Display student details
```

```
dis_stu_det(student)
```

```
56) class Employee:
```

```
    def __init__(self, id=0, name="", address="", age=0, salary=0.0):
```

```
        self.id = id
```

```
        self.name = name
```

```
        self.address = address
```

```
        self.age = age
```

```
        self.salary = salary
```

```
def read_employee_details(emp):
```

```
    print("Enter Employee Details:")
```

```
    emp.id = int(input("ID: "))
```

```
    emp.name = input("Name: ")
```

```
    emp.address = input("Address: ")
```

```
    emp.age = int(input("Age: "))
```

```
    emp.salary = float(input("Salary: "))
```

```
    print()
```

```
def display_employee_details(emp):
```

```
    print("\nEmployee Details:")
```

```
    print(f"ID: {emp.id}")
```

```
    print(f"Name: {emp.name}")
```

```
    print(f"Address: {emp.address}")
```

```
    print(f"Age: {emp.age}")
```

```
    print(f"Salary: {emp.salary:.2f}")
```

```
# Start of the program
```

```
print("\t\t\tWelcome to the Python Program!!\n")
```

```
# Create an Employee object
```

```
employee = Employee()
```

```
# Read employee details
```

```
read_employee_details(employee)
```

Display employee details

display_employee_details(employee)

57) # Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

Input number of elements

n = int(input("Enter the number of elements in the array: "))

Input elements into the array

array = [0] * n

print(f"Enter {n} elements: ")

for i in range(n):

 array[i] = int(input())

Input the element to search

x = int(input("Enter the element to search: "))

Search for the element

found = False

for i in range(n):

 if array[i] == x:

 print(f"Element found at index {i}")

 found = True

 break

if not found: print("Element not found in the array")

58) # Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

Input number of elements

n = int(input("Enter the number of elements in the array: "))

Initialize the list with zeroes

array = [0] * n

Input elements into the list

print(f"Enter {n} elements: ")

for i in range(n):

 array[i] = int(input())

Calculate the sum of all elements

sum_elements = 0

for i in range(n):

```

    sum_elements += array[i]

# Display the sum
print(f"Sum of all elements in the array: {sum_elements}")

59) # Start of the program
print("\t\t\tWelcome to the Python Program!!\n")

# Input number of elements
n = int(input("Enter the number of elements in the array: "))

# Initialize the list with zeroes
array = [0] * n

# Input elements into the list
print(f"Enter {n} elements: ")

for i in range(n):
    array[i] = int(input())

# Initialize the largest and smallest elements
largest = array[0]
smallest = array[0]

# Find the largest and smallest elements
for i in range(1, n):
    if array[i] > largest: largest = array[i]
    if array[i] < smallest: smallest = array[i]

# Display the results
print(f"Largest element in the array: {largest}")
print(f"Smallest element in the array: {smallest}")

60) # Start of the program
print("\t\t\tWelcome to the Python Program!!\n")

# Input number of rows and columns
print("Enter the number of rows and columns of the 2-D array:")

rows = int(input("Number of rows: "))
cols = int(input("Number of columns: "))

# Initialize the 2-D list
array = [[0] * cols for _ in range(rows)]

# Input elements into the 2-D list
print("Enter the elements of the array:")

for i in range(rows):

```

```

    for j in range(cols):
        array[i][j] = int(input(f"Element at [{i}][{j}]: "))

# Input row and column indices to access
print("Enter the row index and column index of the element to access:")
rowIndex = int(input("Row index: "))
colIndex = int(input("Column index: "))

# Check if indices are valid
if 0 <= rowIndex < rows and 0 <= colIndex < cols:
    print(f"Element at [{rowIndex}][{colIndex}]: {array[rowIndex][colIndex]}")
else: print("Invalid indices.")

61) # Start of the program
print("\t\tWELCOME TO THE PYTHON PROGRAM!!\n")

# Dimensions of the matrices
rows, cols = 2, 3

# Initialize matrices
mat1 = [[0] * cols for _ in range(rows)]
mat2 = [[0] * cols for _ in range(rows)]
sum_matrix = [[0] * cols for _ in range(rows)]

# Input Matrix 1 data
print("Enter Matrix 1 Data:")

for i in range(rows):
    for j in range(cols):
        mat1[i][j] = int(input(f"mat1[{i}][{j}]: "))

# Input Matrix 2 data
print("Enter Matrix 2 Data:")

for i in range(rows):
    for j in range(cols):
        mat2[i][j] = int(input(f"mat2[{i}][{j}]: "))

# Calculate the sum of both matrices
print("Sum of the Matrices:")

for i in range(rows):
    for j in range(cols):
        sum_matrix[i][j] = mat1[i][j] + mat2[i][j]

        print(sum_matrix[i][j], end=" ")

```

```

    print()

62) # Start of the program

print("\t\t\tWelcome to the Python Program!!\n")

# Input size of the matrix

n = int(input("Enter the size of the matrix (n): "))

# Initialize the matrix

matrix = [[0] * n for _ in range(n)]

# Input elements of the matrix

print("Enter the elements of the matrix:")

for i in range(n):

    for j in range(n):

        matrix[i][j] = int(input(f"matrix[{i}][{j}]: "))

# Calculate the sum of elements where (i + j) % 2 == 0

sum_elements = 0

for i in range(n):

    for j in range(n):

        if (i + j) % 2 == 0:

            sum_elements += matrix[i][j]

# Print the result

print(f"The sum of elements is: {sum_elements}")

63) import math

def calculate(a, b):

    # Calculate the sum of square roots

    sum_sqrt = math.sqrt(a) + math.sqrt(b)

    # Calculate the product of all numbers from 1 to a

    product = 1

    for i in range(1, int(a) + 1):

        product *= i

    return sum_sqrt, product

def main():

    # Start of the program

    print("\t\t\tWelcome to the Python Program!!\n")

    # Input two numbers A and B

    a = float(input("Enter two numbers A and B: "))

```

```

b = float(input())

# Calculate results

sum_sqrt, product = calculate(a, b)

# Display results

print(f"Sum of square roots of A and B: {sum_sqrt:.2f}")

print(f"Product of all numbers between 1 and A: {product}")

if __name__ == "__main__":

    main()

```

64) def factorial(n):

```

    # Base case: factorial of 0 or 1 is 1

```

```

    if n == 0 or n == 1:

```

```

        return 1

```

```

    else:

```

```

        # Recursive case

```

```

        return n * factorial(n - 1)

```

```

def main():

```

```

    # Start of the program

```

```

    print("\t\t\tWelcome to the Python Program!!\n")

```

```

    # Input a number to calculate its factorial

```

```

    num = int(input("Enter a number to calculate its factorial: "))

```

```

    if num < 0:

```

```

        print("Factorial of a negative number is not defined.")

```

```

    else:

```

```

        print(f"The factorial of {num} is {factorial(num)}")

```

```

if __name__ == "__main__":

```

```

    main()

```

65) # Start of the program

```

print("\t\t\tWelcome to the Python Program!!\n")

```

```

# Read a sentence from the user

```

```

sentence = input("Enter a sentence: ")

```

```

# Display the entered sentence

```

```

print("You have entered:", sentence)

```

66) def count_words(s):

```

    word_count = 0

```

```

in_word = False

for char in s:
    if char in ' \t\n':
        in_word = False
    else:
        if not in_word:
            in_word = True
            word_count += 1

return word_count

def main():
    print("\t\tWelcome to the Python Program!!\n")
    x = input("Enter a string: ")
    total_words = count_words(x)
    print(f"Total number of words: {total_words}")

if __name__ == "__main__":
    main()

67) print("\t\tWelcome to the Python Program!!\n")
# Get input string from the user
A = input("Enter a string of length 10: ")
# Check if the input length is exactly 10 characters
if len(A) != 10:
    print("The string must be exactly 10 characters long.")
else:
    # Extract the last 5 characters
    B = A[5:10]
    # Print the results
    print(f"String A: {A}")
    print(f"The Last 5 copied characters are: {B}")

68) def reverse_string(s):
    """Reverse the given string."""
    return s[::-1]

def main():
    print("\t\tWelcome to the Python Program!!\n")
    # Get input string from the user

```



```

x = input("Enter a string: ")

# Reverse the string using the function
reversed_string = reverse_string(x)

# Print the result
print(f"The Reversed String Is: {reversed_string}")

if __name__ == "__main__":
    main()

69) # Start of the program

# Print welcome message
print("\t\t\tWelcome to the Python Program!!\n")

# Get input string from the user
real = input("Enter a string of exactly 10 characters: ")

# Check if the length is exactly 10 characters
if len(real) != 10:

    print("The input string must be exactly 10 characters long.")
else:

    # Extract substrings
    string1 = real[0:3] # First 3 characters
    string2 = real[3:7] # Characters from position 4 to 7

    # Concatenate substrings
    result = string1 + string2

    # Print results
    print("The Original string is:", real)
    print("The String 1 (positions 1-3) is:", string1)
    print("The String 2 (positions 4-7) is:", string2)
    print("The Concatenated result is:", result)

70) def remove_digits_and_symbols(input_string):

    # Initialize an empty result string
    result = ""

    # Iterate through each character in the input string
    for char in input_string:

        # Check if the character is an alphabet letter
        if char.isalpha():

            result += char

```

```
# Return the result string

return result

def main():

    # Print welcome message

    print("\t\tWelcome to the Python Program!!\n")

    # Get input string from the user

    x = input("Enter a string containing alphabets, digits, and special symbols: ")

    # Call the function to remove digits and special symbols

    filtered_string = remove_digits_and_symbols(x)

    # Print the filtered string

    print("The String after removing digits and special symbols:", filtered_string)

# Run the main function

if __name__ == "__main__":

    main()
```